

Shay Mordechai

Security Researcher | Web Security, Automation & OS Internals Specialist

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Summary

Security Researcher specializing in **Vulnerability Discovery**, **Web Internals**, and **Automation Logic**. Highly skilled in deconstructing complex systems—from **JIT engines** to **Browser Serialization flows**. Recently identified **3 Zero-Day vulnerabilities** in compiler infrastructure (SBCL) using custom AI-augmented tooling. Bringing proven offensive capabilities in **Bot Development** (Predator Project) and **Network Protocol Analysis**, targeting the **Senior Web Security Researcher** role to leverage a unique "builder-breaker" mindset.

Professional Experience

DevSecOps Intern | Israel Tax Authority | 01/2024 – Present

- **Orchestrated** a Defense-in-Depth strategy across the SDLC, integrating **SAST/SCA automation** into **15+ CI/CD pipelines**, reducing vulnerability escape rates by **~30%**.
- **Hardened** high-volume transaction systems by enforcing strict **JSON/XML schema validation** on IBM DataPower, effectively blocking **100%** of malformed payload attacks at the runtime edge.
- **Facilitated** secure system design reviews with **5+ SOC analysts**, translating complex threat models into actionable incident response procedures and reducing Mean Time to Detect (MTTD).

Combat Soldier | Intelligence Gathering Unit, IDF | 11/2017 – 07/2020

- Operated advanced technological systems in a high-pressure operational environment.
- Developed strong analytical capabilities and real-time decision-making skills under uncertainty.

Technical Projects

Automated Trading Bot ("Predator") – Bot Detection & Bypass Research

- **Developed** a high-frequency trading bot using **Python** and **Selenium/Playwright**, implementing advanced **automation logic** to mimic human telemetry.
- **Researched** and bypassed enterprise **Anti-Bot** mechanisms (fingerprinting, behavioral analysis) by randomizing user-agent traits and input timing.
- **Implemented** statistical analysis modules to detect market anomalies in real-time, executing sub-second trade decisions.

AI-Augmented Vulnerability Research – SBCL Compiler (Zero-Days)

- **Engineered** a custom AI-driven structural fuzzer that generated **10,000+** recursive macro structures per hour, successfully bypassing standard compiler hygiene checks.
- **Uncovered 3 Zero-Day vulnerabilities** (Memory Corruption & Stack Exhaustion) within the Macro Expansion engine, achieving a **100% reproduction rate**.
- **Status:** Reported to **MITRE** (Tracking ID: 1977672) and Israel National CERT.

Web Protocol Research – React Server Components (RSC)

- **Conducting deep-dive research** into **React Server Components (RSC)** wire protocols and serialization gadgets.
- **Analyzing** potential **Insecure Deserialization** vectors and logic flaws in the boundary between Client and Server components.

LSaaS Platform & Network Toolset – Zero Trust Architecture

- **Architected** a "Dark Server" environment on AWS, eliminating **100% of public attack surface** using Cloudflare Argo Tunnels and rootless Podman containers.
- **Implemented** custom network stacks (HTTP/DNS/SMTP) writing **2,000+ lines of Python** using **Raw Sockets** to simulate MITM attacks.

Education

B.Sc. Computer Science | Jerusalem College of Technology | **2021-2025** | GPA: 89

Relevant coursework: Network Analysis (97), Reverse Engineering (93), Information Security (90)

Technical Skills

- **Security Research:** Web Internals, **Browser Automation**, JIT Internals, Memory Corruption, Zero-Day Discovery, Fuzzing.
- **Scripting & Automation:** Expert in **Python**, **Bash**, **JavaScript** (Node.js/React), Selenium, Playwright.
- **Operating Systems:** Daily **Fedora Kinoite** user; Deep understanding of **Linux Internals** (Namespaces, Cgroups, eBPF).
- **Network Analysis:** L2-L7 protocol research, Raw Sockets, Scapy, Wireshark, Traffic Parsing.
- **Tools:** IDA Pro, Burp Suite, Docker/Podman, AWS (Zero Trust), Git.